

# **Introduction to Drone Technology**

## **Module 2 – Global Navigation Satellite Systems (GNSS)**

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# 3 Global Navigation Satellite Systems – GNSS

## Global Systems

- **GPS** (Global Positioning System): USA, 24+ satellites; the original and best-known GNSS
- **BeiDou** (BDS): China; 35 satellites; high orbits; integrates navigation and communication functions
- **Galileo**: European Union; 24+ satellites, only civilian-controlled GNSS
- **GLONASS**: Russia; 24+ satellites; fully operational with global coverage

## Regional Systems (RNSS)

- **NavIC**: India; 7 satellites; autonomous system covering India and 1500 km around the mainland
- **QZSS**: Japan; 7+ satellites; complements GPS in East Asia and Oceania

## 3.1 GPS Architecture

### Space Segment

The space segment is a group of satellites that orbit the Earth at an altitude of 20000 km. Each satellite transmits a one-way radio signal that contains its position and current time. At least 4 satellites must be visible from any point on Earth to provide global coverage.

### Control Segment

The control segment is a global network of monitoring and ground control stations. Its objective is the maintenance of satellite orbits, clock adjustment and health and status monitoring.

### User Segment

The user segment comprises all applications on Earth that use GPS services with appropriate receiver equipment for navigation. The receiver collects the transmitted information from visible satellites and calculates its position.

## 3.2 GNSS Error Sources

### Clock-Related Errors

Although satellite clocks are very precise, they have a small drift that causes an error of up to 5.2 m per day. The drift of the cheaper internal receiver clock is significantly higher and is treated as a 4th unknown. Therefore, 4 visible satellites are needed to calculate the position [1].

### Sagnac Effect

The Sagnac effect is a relativistic error caused by Earth's rotation during signal propagation, resulting in an error magnitude of 30 m if uncompensated [1].

### Ionosphere Delay

At an altitude of about 1000 km, the signal passes through the ionosphere, which consists of ionized gases. Because of its dispersive properties, the delay is frequency dependent. The error reaches up to 100 m under severe conditions [1].

### Troposphere Delay

The troposphere, the lowest atmosphere layer up to 20 km, causes a delay that is complex to model because of the wet component. The delay is frequency independent and cannot be compensated by using two frequencies. The error lies in the range of 2.5 to 25 m [1].

## **Multipath Errors**

Surrounding structures such as buildings or trees reflect satellite signals, so that they reach the receiver via multiple paths. The delayed echoes are superimposed on the direct signal. The error magnitude can reach up to 100 m depending on the environment [1].

## **Intentional Errors**

Intentional errors caused by a third party with the objective of hindering or manipulating GNSS usage. Jamming is the broadcast of a signal at the same frequency and higher power near the target receiver. Spoofing uses a fake signal designed to trick the receiver into believing that it is a authentic signal [1].

## **3.3 Dilution of Precision (DOP)**

- **GDOP** (Geometric DOP): General term describing satellite geometry that is used as an overall accuracy factor. Lower DOP means more optimal geometry and better accuracy.
- **PDOP** (Position DOP): Geometry factor for three dimensional position accuracy.
- **HDOP** (Horizontal DOP): Geometry factor for horizontal position accuracy.
- **VDOP** (Vertical DOP): geometry factor for vertical position accuracy.

## **3.4 Real Time Kinematics (RTK) GNSS**

RTK is a differential GNSS technique based on carrier phase measurements. The distance is obtained by determining the number of complete carrier cycles of the signal. The unknown number of complete cycles is called integer ambiguity. Additionally, the information provided by a reference station is used to calculate the relative position and the position of the station is then added to calculate the absolute position in a global coordinate frame.

## **3.5 GNSS Accuracy**

### **1. Standard Positioning Service (SPS)**

SPS is a GPS service available that provides access to L1, L2 and L5 frequencies for civilian users. The expected accuracy is  $< 8$  m horizontally and  $< 13$  m vertically with 95 % probability [2].

### **2. Differential GPS (DGPS)**

DGPS improves primary GNSS information by the use of a network of ground based reference stations which enable the broadcasting of differential information to improve the accuracy. The horizontal accuracy is 1-2 m in the service region [3].

### **3. RTK Float**

RTK float is a differential GNSS technique that uses carrier phase measurements with the integer ambiguities estimated as floating point numbers. The true integer number of complete wavelengths has not been resolved. The accuracy is at the meter level  $< 1$  m [4].

### **4. RTK Fixed**

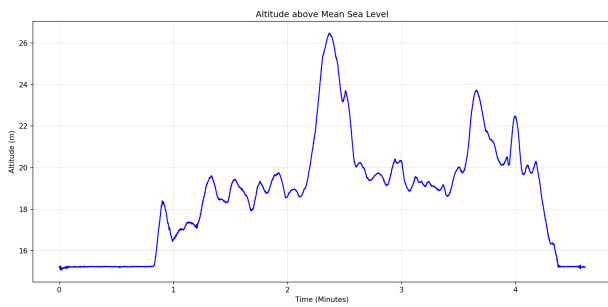
RTK fixed is a differential GNSS technique that uses carrier phase measurements with the integer ambiguities resolved. The accuracy is  $< 5$  cm [4].

## 4 Coordinate Systems

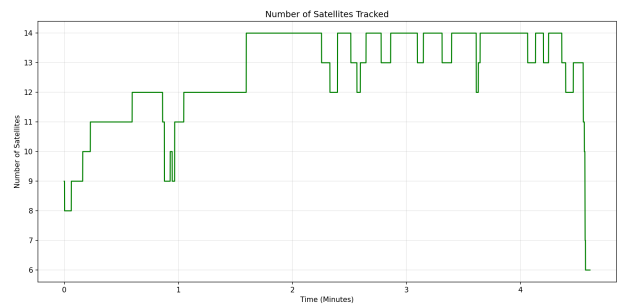
### 4.1 Universal Transverse Mercator (UTM) Accuracy

- Reference position used: N55.47° E010.33°
- Error in latitude for a 1000 m step: 1.594390 m
- Error in longitude for a 1000 m step: 3.750405 m
- The leading zero in “010.33” is used because longitude values range from 0° to 180°

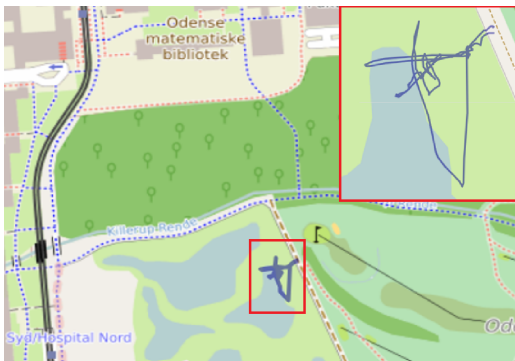
### 4.2 National Marine Electronics Association (NMEA) 0183 Data



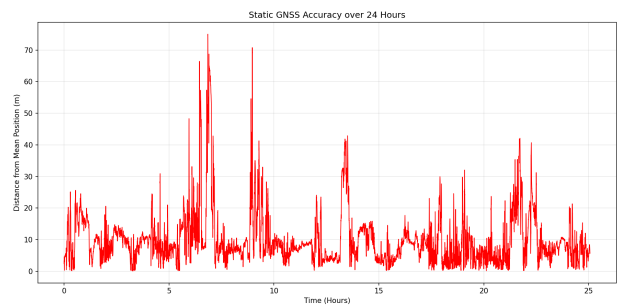
(a) Altitude above MSL vs. time.



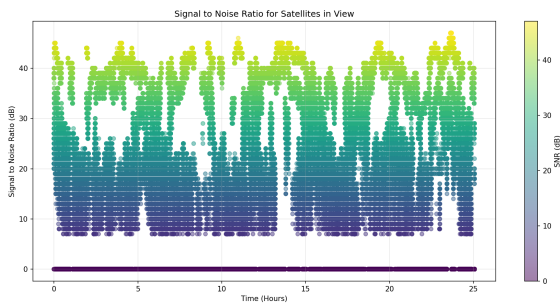
(b) Number of satellites tracked vs. time.



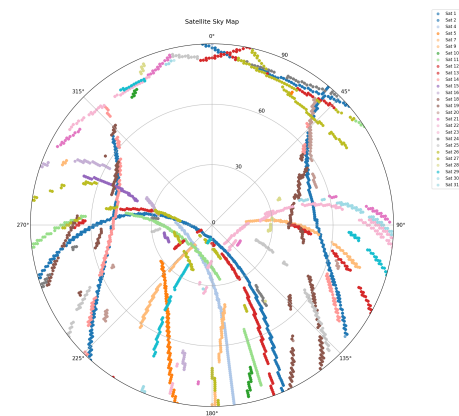
(c) Drone track during the flight.



(d) Static GNSS accuracy over 24 hours.



(e) SNR for satellites in view over 24 hours.



(f) Hemisphere map of satellite locations over 24 hours.

Figure 1: NMEA 0183 data analysis for the drone flight and the 24-hour static test.

## References

- [1] M. Karaim, M. Elsheikh, and A. Noureldin, “GNSS Error Sources,” in *Multifunctional Operation and Application of GPS*, Rijeka, Croatia: InTech, 2018, pp. 69–85, doi: 10.5772/intechopen.75493.
- [2] U.S. Department of Defense, “Global Positioning System Standard Positioning Service Performance Standard,” 5th ed., Apr. 2020. [Online]. Available: <https://www.gps.gov/sites/default/files/2025-07/2020-SPS-performance-standard.pdf>
- [3] U.S. Department of Defense, U.S. Department of Homeland Security, and U.S. Department of Transportation, “2019 Federal Radionavigation Plan,” Rep. DOT-VNTSC-OST-R-15-01, 2019. [Online]. Available: <https://navcen.uscg.gov/sites/default/files/pdf/FederalRadionavigationPlan2019.pdf>
- [4] D. S. M. Valente, A. Momin, T. Grift, and A. Hansen, “Accuracy and precision evaluation of two low-cost RTK global navigation satellite systems,” *Computers and Electronics in Agriculture*, vol. 168, p. 105142, Jan. 2020, doi: 10.1016/j.compag.2019.105142.